



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2012

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 180 minutes
- Write using black pen.
- Board approved calculators may be used
- Show all necessary working in Questions 11–16
- A table of standard integrals is on the back of the multiple choice answer sheet

Total Marks - 100 Marks

Section I 10 Marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II 90 Marks

- Attempt Questions 11–16
- Allow about 2 hour 45 minutes for this section.

Examiner: *External Examiner*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

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Section I

Objective-response Questions

Total marks – 10

Attempt Questions 1 – 10

Answer each question on the multiple choice answer sheet provided.

- 1 Let $u = 7 \cos \frac{\pi}{4} + 7i \sin \frac{\pi}{4}$ and $v = a \cos b + ai \sin b$, where a and b are real constants.

If $uv = 42 \cos \frac{\pi}{20} + 42i \sin \frac{\pi}{20}$, then

- (A) $a = 35$ and $b = \frac{\pi}{5}$ (B) $a = 6$ and $b = \frac{\pi}{5}$
 (C) $a = 35$ and $b = -\frac{\pi}{5}$ (D) $a = 6$ and $b = -\frac{\pi}{5}$

- 2 If $z^2 = 4 \operatorname{cis} \left(\frac{4\pi}{3} \right)$, then z is equal to

- (A) $\sqrt{3} + i$ or $-\sqrt{3} - i$ (B) $1 - \sqrt{3}i$ or $-1 + \sqrt{3}i$
 (C) $\sqrt{3} - i$ or $\sqrt{3} + i$ (D) $1 - \sqrt{3}i$ or $1 + \sqrt{3}i$

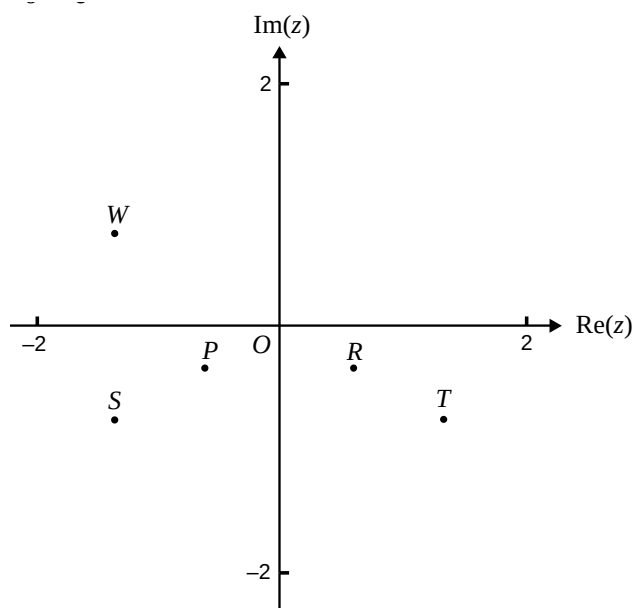
- 3 Let $z = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}$.

The imaginary part of $z - i$ is

- (A) $-\frac{i}{2}$ (B) $-\frac{3i}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

- 4 The point W on the Argand diagram below represents a number w where $|w| = 1.5$.
 The number w^{-1} is best represented by the point

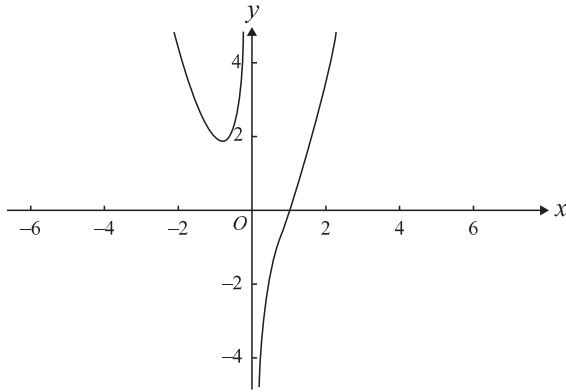
- (A) P (B) R (C) S (D) T



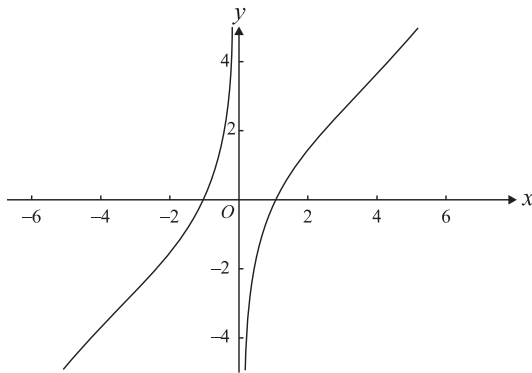
- 5 $P(z)$ is a polynomial in z of degree 4 with real coefficients
Which one of the following statements must be false?
- (A) $P(z)$ has four real roots.
 (B) $P(z)$ has two real roots and two non-real roots.
 (C) $P(z)$ has one real root and three non-real roots.
 (D) $P(z)$ has no real roots.
- 6 The graph of $f(x) = \frac{1}{x^2 + mx - n}$, where m and n are real constants, has no vertical asymptotes if
- (A) $m^2 < -4n$ (B) $m^2 > -4n$ (C) $m^2 < 4n$ (D) $m^2 > 4n$
- 7 Consider the graph of $f(x) = \sin^3 x$ for $-\pi \leq x \leq 2\pi$.
The area bounded by the graph of $f(x)$ and the x -axis could be found by evaluating
- (A) $\int_{-1}^1 (1-u^2) du$ (B) $3 \int_{-1}^1 (1-u^2) du$
 (C) $-\int_{-1}^1 (1-u^2) du$ (D) $-3 \int_{-1}^1 (1-u^2) du$
- 8 Given that $\frac{dy}{dx} = y^2 + 1$, and that $y = 1$ at $x = 0$, then
- (A) $y = y^2 x + x + 1$ (B) $y = \tan\left(x + \frac{\pi}{4}\right)$
 (C) $y = \tan\left(x - \frac{\pi}{4}\right)$ (D) $x = \log_e\left(\frac{y^2 + 1}{2}\right)$
- 9 The velocity v m/s of a body which is moving in a straight line, when it is x m from the origin, is given by $v = \sin^{-1} x$.
The acceleration of the body in m/s^2 is given by
- (A) $-\cos^{-1} x$ (B) $\cos^{-1} x$ (C) $-\frac{\sin^{-1} x}{\sqrt{1-x^2}}$ (D) $\frac{\sin^{-1} x}{\sqrt{1-x^2}}$

- 10 Let $f(x) = \frac{x^k + a}{x}$, where k and a are real constants.

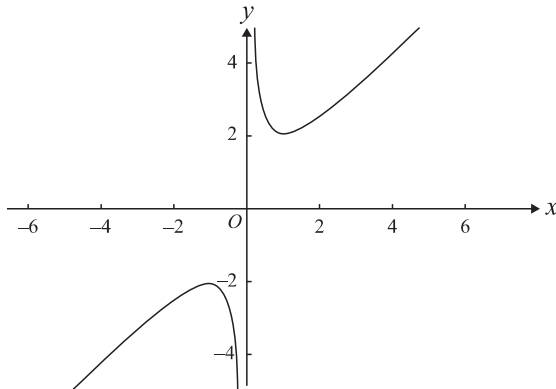
If k is an odd integer which is greater than 1 and $a < 0$, a possible graph of f could be
(A)



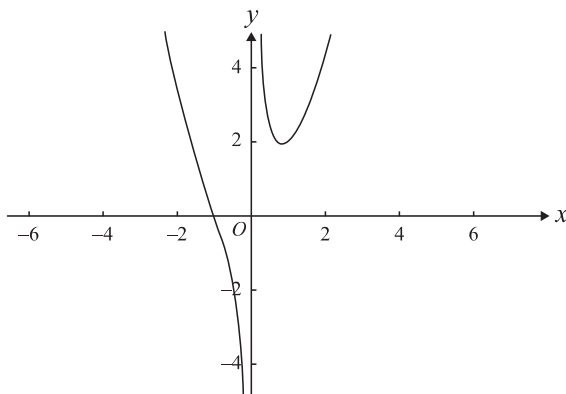
(B)



(C)



(D)



End of Section I

Section II**Free response Questions****Total marks – 90****Attempt Questions 11 – 16**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Use the substitution $x = \sin^2 \theta$ to evaluate $\int_0^{\frac{1}{2}} \frac{\sqrt{x}}{(1-x)^{\frac{3}{2}}} dx$ 3

(b) Find $\int x\sqrt{3-x} dx$. 2

(c) (i) By completing the square, find the exact value of $\int_{\frac{1}{8}}^{\frac{1}{4}} \frac{1}{\sqrt{2x(1-2x)}} dx$ 2

(ii) Hence, evaluate $\int_{\frac{1}{8}}^{\frac{1}{4}} \frac{1-x}{\sqrt{2x(1-2x)}} dx$ 2

(d) Find the value of the discriminant for the quadratic equation $(1+i)z^2 + 4iz - 2(1-i) = 0$ 2

(e) (i) Find the value of $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6$. 1

(ii) Show that $(\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta) = 1 + \cos \theta + i \sin \theta$. 1

(iii) Hence show that $\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 = 0$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The line $x = 8$ is a directrix of the ellipse with equation 2

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0$$

and $(2, 0)$ is the corresponding focus.
Find the value of a and b .

- (b) (i) Show that $2 - i$ is a solution of the equation $z^3 - (2 - i)z^2 + z - 2 + i = 0$. 2

- (ii) Hence find all the solutions of the equation $z^3 - (2 - i)z^2 + z - 2 + i = 0$. 2

- (c) Consider the function $f(x) = \log_e(4 - x^2)$.

- (i) By first sketching $y = 4 - x^2$, sketch $y = f(x)$. 2

Let A be the magnitude of the area enclosed by the graph of $y = f(x)$,
the coordinate axes and the line $x = 1$.

- (ii) Without evaluating A , use (i) to show that $\log_e 3 < A < \log_e 4$. 1

- (iii) Find $\int \frac{x^2}{4 - x^2} dx$. 3

- (iv) Hence find the exact value of A in the form $a + b \log_e c$,
where a , b and c are integers. 3

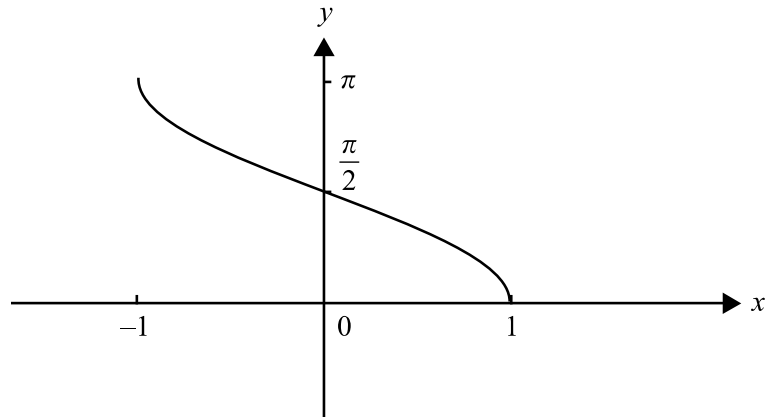
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Prove using induction for integers $n \geq 2$.

3

$$n^{n+1} > n(n+1)^{n-1}$$

- (b) The diagram below shows the graph of $y = \cos^{-1} x$.



Using the method of cylindrical shells, find the exact volume formed if the graph above is rotated about the y -axis.

3

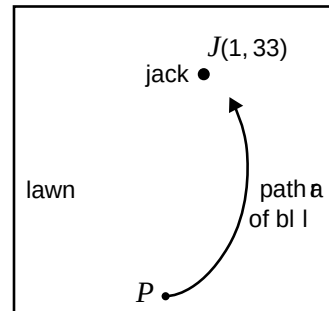
Question 13 continues on the next page

Question 13 continued

- (c) The game of lawn bowls is played on a horizontal lawn. The aim is to roll a ball (usually called a ‘bowl’) to come to rest as close as possible to a target ball called the ‘jack’.



Bowler



View from above

All displacements are in metres.

At one stage during the game, the jack is at the point $J(1, 33)$.

The path of a particular ball in this game is modelled by:

$$x = 2 \sin\left(\frac{2t}{15}\right) \text{ and } y = 2 + \frac{5}{3}t - \frac{5}{3}\sin\left(\frac{t}{3}\right), \quad 0 \leq t \leq \frac{15\pi}{2}$$

where t is the time in seconds after the ball is released from the point P .

- | | | |
|-------|--|---|
| (i) | Write down the coordinates of P . | 1 |
| (ii) | Find expressions for the components of velocity, in metres per second, of the ball at time t seconds after the ball is released. | 2 |
| (iii) | At the instant the ball is released, what angle does its path make with the forward direction?
Give your answer correct to 1 decimal place. | 2 |
| (iv) | At what time, correct to the nearest tenth of a second, does the ball begin to swing left towards the jack? | 2 |
| (v) | Determine whether the path of the ball passes through J . | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

A ‘parasailing’ water-skier i.e. a water-skier with a parachute attached of mass 90 kg is towed by a boat in a straight line from rest.

The boat exerts a constant force of 410 N acting horizontally on the skier.

At this stage the resistance acting on the skier is a constant 50 N, which acts horizontally.

- (a) By use of a force diagram, show that the acceleration of the skier is 4 m/s^2 . 2
- (b) By starting with $a = 4$, show that the speed of the skier, is given by $v^2 = 8x$, where x is the horizontal distance travelled by the skier. 2
Hence show that having been towed a distance of 32 m, his speed is 16 m/s.

After the skier has been towed 32 m across the water the drag of the parachute becomes significant. The drag of the parachute produces an *additional* resistance of $6v \text{ N}$ to the horizontal motion of the skier, where $v \text{ m/s}$ is the velocity of the skier. Let $a \text{ m/s}^2$ is the acceleration of the skier.

- (c) Show that $a = \frac{1}{15}(60 - v)$ 1
- (d) Find the time required to reach a speed of 20 m/s from a speed of 16 m/s. 3
Give your answer in seconds, correct to one decimal place.

After some time, the parasailing skier is being towed horizontally at a *constant speed* and at a fixed distance above the water.

The tow rope from the boat makes an angle of 30° to the horizontal, and the parachute cord makes an angle of θ to the horizontal.

The diagram below shows all the forces that are now acting on the parasailing water skier:

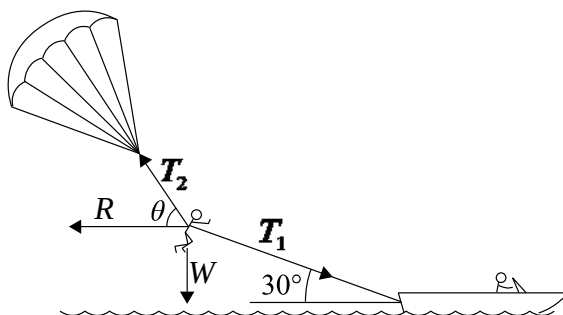
The tow rope now exerts a force, T_1 , of 500 N on the skier.

The skier is experiencing a horizontal resistance, R , of 100 N.

Let the tension exerted by the parachute cord on the skier be T_2 ,

and the force due to gravity on the skier be W .

Take $g = 10$, where g is the magnitude of the acceleration due to gravity.



- (e) By resolving in the horizontal and vertical directions, show that 3
- $$\begin{cases} 500 \cos 30^\circ - T_2 \cos \theta - 100 = 0 \\ T_2 \sin \theta - 500 \sin 30^\circ - 90g = 0 \end{cases}$$

Question 14 continues on the next page

Question 14 continued

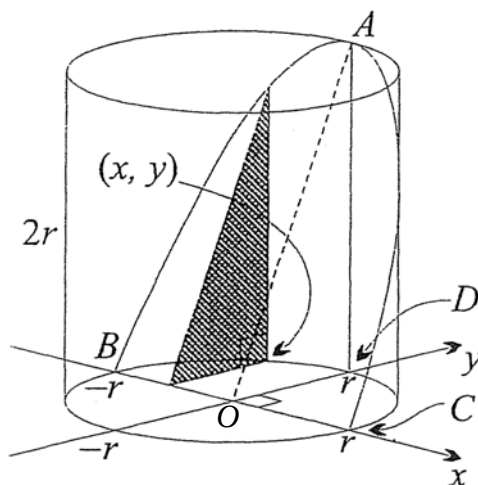
(f) Show that $\tan \theta = \frac{115}{25\sqrt{3} - 10}$. 2

(g) Hence, find the value of T_2 correct to the nearest integer. 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows a cylindrical wedge $ABCD$, the cross sections of which are all right triangles. Each cross section is similar to triangle AOD . The base of each cross section is parallel to OD . The height of the cylinder is equal to the diameter of its base. Let the radius of the base be r units.



- (i) Show that the typical triangular cross-section shaded has area $(r^2 - x^2)$ square units. 2
- (ii) Hence find the volume of the wedge. 2
- (b) For positive real numbers x and y
- (i) Prove that $\frac{x+y}{2} \geq \sqrt{xy}$ 2
When is there equality?
- (ii) Hence by considering $\frac{1}{a} + \frac{1}{b}$, or otherwise, prove that $\frac{2ab}{a+b} \leq \sqrt{ab}$ 1
for positive real numbers a, b .
- (iii) Hence, or otherwise prove that $\frac{1}{x-1} + \frac{1}{x} + \frac{1}{x+1} > \frac{3}{x}$ for any $x > 1$ 2
- (iv) If $H = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \dots + \frac{1}{n}$, where n is an integer $n > 1$, 2
use (iii) to show that $\lim_{n \rightarrow \infty} H = \infty$.

Question 15 continues on the next page

Question 15 continued

- (c) (i) Given that ω is one of the non-real roots of $z^3 = 1$, **1**
show that $1 + \omega + \omega^2 = 0$.

- (ii) Using (i), or otherwise, show that **3**

$$\left(\frac{\omega}{1+\omega}\right)^k + \left(\frac{\omega^2}{1+\omega^2}\right)^k = (-1)^k 2 \cos \frac{2}{3} k \pi, \text{ where } k \in \mathbb{Z}.$$

End of Question 15

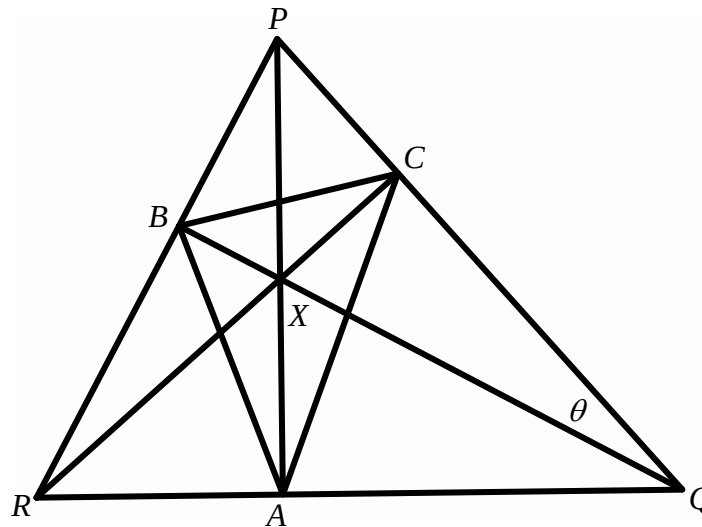
Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) $I_n = \int_0^a (a-x)^n \cos x \, dx$, $a > 0$ and n is an integer with $n \geq 0$.

(i) Show that, for $n \geq 2$, $I_n = na^{n-1} - n(n-1)I_{n-2}$. 3

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right)^3 \cos x \, dx$ 3

- (b) In the figure below, $\triangle PQR$ is acute angled and AP , BQ and CR are altitudes concurrent at X . Also $\angle XQC = \theta$. $\triangle ABC$ is called the *pedal triangle* of $\triangle PQR$.



- (i) Prove that $\angle XRB = \theta$. 2
- (ii) Prove that X , A , Q and C are concyclic. 1
- (iii) Deduce that $\angle XAC = \theta$. 1
- (iv) Hence deduce that in an acute angled triangle the altitudes bisect the angles of the pedal triangle through which they pass. 2

Question 16 continues on the next page

Question 16 continued

- (c) (i) A binary string is a sequence of **1**s and **0**s,
e.g. **1 1 0 1 1 1 1 0 0 1 0 1** is a binary string of length 12.

In a binary string of length 50, how many ways are there to
have a string with exactly 9 **1**s and that no two **1**s are adjacent?
Justify your answer.

2

- (ii) Given 50 cards with the integers 1, 2, 3, ... 50 printed on them,
how many ways are there to select 9 distinct cards, such that no
two cards have consecutive numbers printed on them?
(An answer with no reasoning will get no credit.)

1

End of paper

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Student Number: _____

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Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct (arrow pointing to B)

Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. A ☐ B ☐ C ☐ D ☒
2. A ☐ B ☒ C ☐ D ☐
3. A ☐ B ☐ C ☒ D ☐
4. A ☒ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☒ D ☐
6. A ☒ B ☐ C ☐ D ☐
7. A ☐ B ☒ C ☐ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☒
10. A ☒ B ☐ C ☐ D ☐

Q11 (a) $I = \int_0^1 \frac{\sqrt{x}}{(1-x)^{3/2}} dx$ let $x = \sin^2 \theta$
 $dx = 2 \sin \theta \cos \theta d\theta$

$$= \int_0^{\pi/4} \frac{\sin \theta}{(1 - \sin^2 \theta)^{3/2}} \cdot 2 \sin \theta \cos \theta d\theta$$

$$= 2 \int_0^{\pi/4} \frac{\sin^2 \theta \cos \theta d\theta}{\cos^3 \theta}$$

$$= 2 \int_0^{\pi/4} \tan^2 \theta \cdot d\theta$$

$$= 2 \cdot \int_0^{\pi/4} (\sec^2 \theta - 1) d\theta$$

$$= 2 \left[\tan \theta - \theta \right]_0^{\pi/4}$$

$$= 2 \left[1 - \frac{\pi}{4} \right]$$

$$= \underline{\underline{\left| 2 - \frac{\pi}{2} \right|}}$$

(b) $\int x \sqrt{3-x} \cdot dx$ let $u = 3-x$
 $\therefore x = 3-u$
 $dx = -du$

$$= - \int (3-u) \sqrt{u} \cdot du$$

$$= \int (u^{3/2} - 3u^{1/2}) du$$

$$= \frac{2}{5} u^{5/2} - 2 u^{3/2} + c$$

$$= \underline{\underline{\left| \frac{2}{5} (3-x)^{5/2} - 2 (3-x)^{3/2} + c \right|}}$$

Q11 (contd)

$\leq (1)$

$$\int_{\frac{1}{8}}^{\frac{1}{4}} \frac{1}{\sqrt{2x-4x^2}} dx.$$

$$= \frac{1}{2} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{dx}{\sqrt{\frac{x}{2} - x^2}}.$$

$$= \frac{1}{2} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{dx}{\sqrt{\frac{1}{16} - (x - \frac{1}{4})^2}}$$

$$= \frac{1}{2} \left[\sin^{-1} \frac{x - \frac{1}{4}}{\frac{1}{4}} \right]_{\frac{1}{8}}^{\frac{1}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1}(4x-1) \right]_{\frac{1}{8}}^{\frac{1}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1} 0 - \sin^{-1}(-\frac{1}{2}) \right]$$

$$= \frac{1}{2} (0 - -\frac{\pi}{6})$$

$$= \frac{\pi}{12}.$$

$$(11) \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{1-x}{\sqrt{2x-4x^2}} dx = \frac{1}{8} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{8-8x}{\sqrt{2x-4x^2}} dx$$

$$= \frac{1}{8} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{6+2-8x}{\sqrt{2x-4x^2}} dx.$$

$$= \frac{1}{8} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{6}{\sqrt{2x-4x^2}} dx + \frac{1}{8} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{2-8x}{\sqrt{2x-4x^2}} dx.$$

$$= \frac{3}{4} \int_{\frac{1}{8}}^{\frac{1}{4}} \frac{dx}{\sqrt{2x-4x^2}} + \frac{1}{4} \left[\sqrt{2x-4x^2} \right]_{\frac{1}{8}}^{\frac{1}{4}}$$

$$= \frac{3}{4} \times \frac{\pi}{12} + \frac{1}{4} \left[\sqrt{\frac{1}{4}} - \sqrt{\frac{3}{16}} \right]$$

$$= \frac{\pi}{16} + \frac{1}{4} \left(\frac{1}{2} - \frac{\sqrt{3}}{4} \right)$$

$$= \frac{\pi}{16} + \frac{1}{8} - \frac{\sqrt{3}}{16}.$$

Q11 CONTD

$$\begin{aligned} \text{(d)} \quad \Delta &= 16i^2 + 8(1 - i^2) \\ &= -16 + 8(2) \\ &= 0. \end{aligned}$$

$$\begin{aligned} \text{(e)} \quad (i) \quad & \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^6 \\ &= \cos \pi + i \sin \pi \\ &= -1. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \text{LHS} &= (\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta) \\ &= \text{cis } \theta (1 + \text{cis } (-\theta)) \\ &= \text{cis } \theta + \text{cis } 0 \\ &= \text{cis } \theta + 1 \\ &= 1 + \cos \theta + i \sin \theta \\ &= \text{RHS}. \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad & \left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^6 \\ &= \left[\text{cis } \frac{\pi}{6} (1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}) \right]^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^6 \\ &= -1 \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)^6 \\ &= 0 \quad \text{as required} \end{aligned}$$

2012 Extension 2 Mathematics THSC:
Solutions— Question 12

Marks

Question 12 (15 marks)

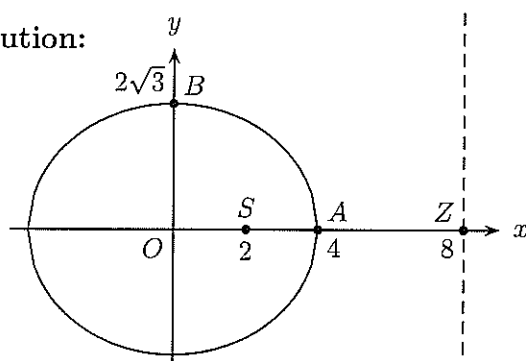
- (a) The line $x = 8$ is a directrix of the ellipse with equation

2

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0$$

and $(2, 0)$ is the corresponding focus.
Find the value of a and b .

Solution:



$$\begin{aligned} ae &= 2 \quad \dots \quad \boxed{1} \\ \frac{a}{e} &= 8 \quad \dots \quad \boxed{2} \\ \boxed{1} \times \boxed{2}: \quad a^2 &= 16, \\ \therefore a &= 4. \\ \frac{SA}{AZ} &= e, \\ &= \frac{2}{4}, \\ &= \frac{1}{2}. \\ b^2 &= a^2(1 - e^2), \\ &= 16(1 - \frac{1}{4}), \\ &= 12, \\ \therefore b &= 2\sqrt{3}. \end{aligned}$$

- (b) (i) Show that $2 - i$ is a solution of the equation $z^3 - (2 - i)z^2 + z - 2 + i = 0$.

2

Solution: L.H.S. $= (2 - i)^3 - (2 - i)(2 - i)^2 + (2 - i) - (2 - i),$
 $= 0,$
 $= \text{R.H.S.}$
 $\therefore (2 - i)$ is a solution.

- (ii) Hence find all the solutions of the equation $z^3 - (2 - i)z^2 + z - 2 + i = 0$.

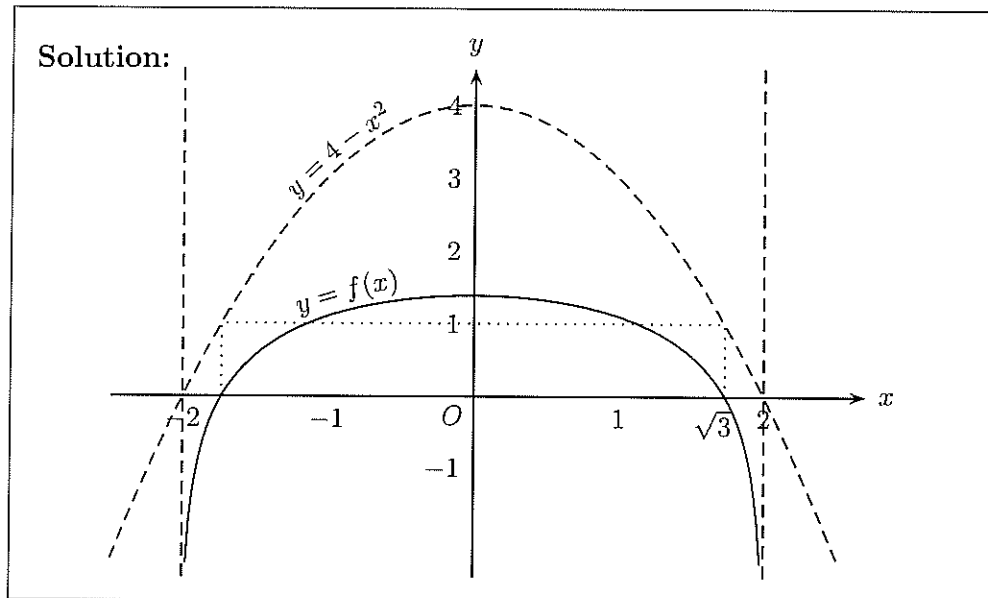
2

Solution: $z(z^2 + 1) - (2 - i)(z^2 + 1) = 0,$
 $(z - 2 + i)(z^2 + 1) = 0,$
 $(z - 2 + i)(z + i)(z - i) = 0.$
 $\therefore$ Solutions $2 - i, \pm i.$

(c) Consider the function $f(x) = \log_e(4 - x^2)$.

(i) By first sketching $y = 4 - x^2$, sketch $y = f(x)$.

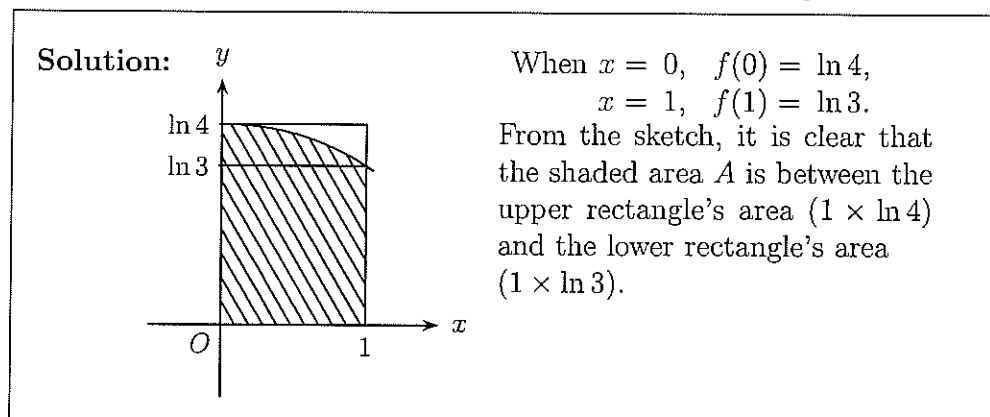
2



Let A be the magnitude of the area enclosed by the graph of $y = f(x)$, the coordinate axes and the line $x = 1$.

(ii) Without evaluating A , use (i) to show that $\log_e 3 < A < \log_e 4$.

1



(iii) Find $\int \frac{x^2}{4-x^2} dx$.

Solution: Method 1—

$$\begin{aligned} I &= \int \frac{x^2 - 4 + 4}{4 - x^2} dx, \\ &= - \int dx + \int \frac{4}{4 - x^2} dx, \\ &= -x + \int \frac{dx}{2 - x} + \int \frac{dx}{2 + x}, \\ &= -x - \ln(2 - x) + \ln(2 + x) + c, \\ &= \ln\left(\frac{2 + x}{2 - x}\right) - x + c. \end{aligned}$$

$$\begin{aligned} \frac{4}{4 - x^2} &\equiv \frac{A}{2 - x} + \frac{B}{2 + x}, \\ 4 &\equiv A(2 + x) + B(2 - x), \\ \text{put } x &= -2, \quad B = 1, \\ x &= 2, \quad A = 1. \end{aligned}$$

Solution: Method 2—

$$\begin{aligned} I &= \int \frac{4 \sin^2 \theta \cdot 2 \cos \theta d\theta}{4 - 4 \sin^2 \theta}, \\ &= \int \frac{2 \sin^2 \theta \cdot 2 \cos \theta}{\cos^2 \theta}, \\ &= 2 \int \frac{1 - \cos^2 \theta}{\cos \theta} d\theta, \\ &= 2 \int (\sec \theta - \cos \theta) d\theta, \\ &= 2 \{ \ln(\sec \theta + \tan \theta) - \sin \theta \} + c, \\ &= 2 \ln \left(\frac{2}{\sqrt{4 - x^2}} + \frac{x}{2} \cdot \frac{2}{\sqrt{4 - x^2}} \right) - x + c, \\ &= 2 \ln \left(\frac{x + 2}{\sqrt{(2 + x)(2 - x)}} \right) - x + c, \\ &= \ln \left(\frac{2 + x}{2 - x} \right) - x + c. \end{aligned}$$

$$\begin{aligned} \text{Put } x &= 2 \sin \theta, \\ dx &= 2 \cos \theta d\theta, \\ \frac{x^2}{4} &= \sin^2 \theta, \\ \frac{4 - x^2}{4} &= \cos^2 \theta. \end{aligned}$$

(iv) Hence find the exact value of A in the form $a + b \log_e c$, where a , b and c are integers.

Solution:

$$\begin{aligned} I &= \int_0^1 \ln(4 - x^2) dx, \\ &= x \ln(4 - x^2) \Big|_0^1 + 2 \int_0^1 \frac{x^2 dx}{4 - x^2}, \\ &= \{\ln 3 - 0\} + 2 \left[\ln \left(\frac{2 + x}{2 - x} \right) - x \right]_0^1, \\ &= \ln 3 + 2 \{ \ln 3 - 1 - (\ln 1 - 0) \}, \\ &= 3 \ln 3 - 2. \end{aligned}$$

$$\begin{aligned} u &= \ln(4 - x^2), \quad v' = 1, \\ u' &= \frac{-2x}{4 - x^2}, \quad v = x. \end{aligned}$$

15 Ex 2

13. (a) Prove $n^{n+1} > n(n+1)^{n-1}$ for $n \geq 2$.

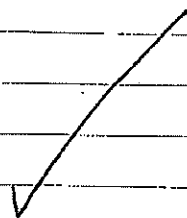
Let $n=2$ LHS = $2^3 = 8$ RHS = $2(3)^1 = 6$
 $\therefore$ true for $n=2$.

Assume true for $n=k$

$$k^{k+1} > k(k+1)^{k-1}$$

or $k^k > (k+1)^{k-1} \quad *$

or $\frac{k^k}{(k+1)^{k-1}} > 1.$



Let $n=k+1$ RTP $(k+1)^{k+1} > (k+2)^k$

OR $\frac{(k+1)^{k+1}}{(k+2)^k} > 1$

Now LHS = $\frac{(k+1)^{k+1}}{(k+2)^k}$

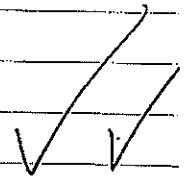
$$> \frac{(k+1)^{k+1}}{(k+2)^k} \times \frac{(k+1)^{k-1}}{k^k}$$

since $\frac{(k+1)^{k-1}}{k^k} < 1$
 from assumption.

$$> \frac{(k+1)^{2k}}{[k(k+2)]^k}$$

$$= \frac{[(k+1)^2]^k}{[k(k+2)]^k}$$

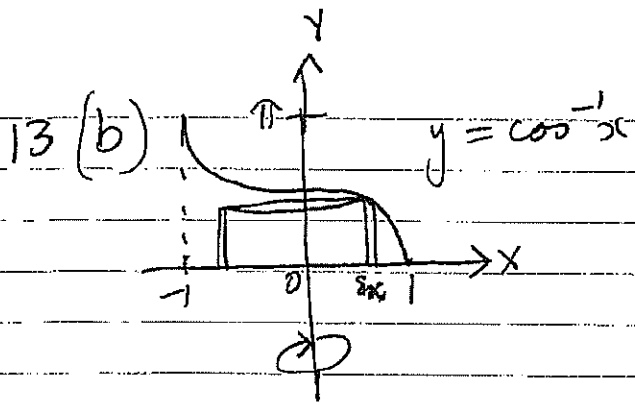
$$= \left[\frac{k^2 + 2k + 1}{k^2 + 2k} \right]^k > 1 \quad \#$$



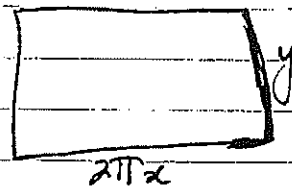
$\therefore$ true for $n=k+1$.

$\therefore$ By P.O.M.I, true $\forall n \geq 2$.

(3)



Area of cylindrical slice = $2\pi xy$



Vol. of cyl. slice = $2\pi xy \delta x$.

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-1}^1 2\pi xy \delta x$$

$$= 2\pi \int_{-1}^1 x \cos^{-1} x \, dx$$

$$V = 4\pi \int_0^1 x \cos^{-1} x \, dx \quad (\text{symmetric}) \quad \begin{array}{l} u = \cos^{-1} x \quad dv = x \\ du = \frac{-1}{\sqrt{1-x^2}} \quad v = \frac{x^2}{2} \end{array}$$

$$= 4\pi \left[\left[\frac{x^2}{2} \cos^{-1} x \right]_0^1 - \int_0^1 \frac{x^2}{2} \left(\frac{-1}{\sqrt{1-x^2}} \right) dx \right]$$

$$= 2\pi \left[x^2 \cos^{-1} x \right]_0^1 + 2\pi \int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx$$

$$= 2\pi \left[x^2 \cos^{-1} x \right]_0^1 + 2\pi \int_0^{\frac{\pi}{2}} \frac{\sin^2 \theta}{\cos \theta} \cos \theta d\theta \quad \begin{array}{l} \text{let } x = \sin \theta \\ dx = \cos \theta d\theta \\ x=0, \theta=0 \\ x=1, \theta=\frac{\pi}{2} \end{array}$$

$$= 2\pi \left[x^2 \cos^{-1} x \right]_0^1 + 2\pi \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2\theta) d\theta$$

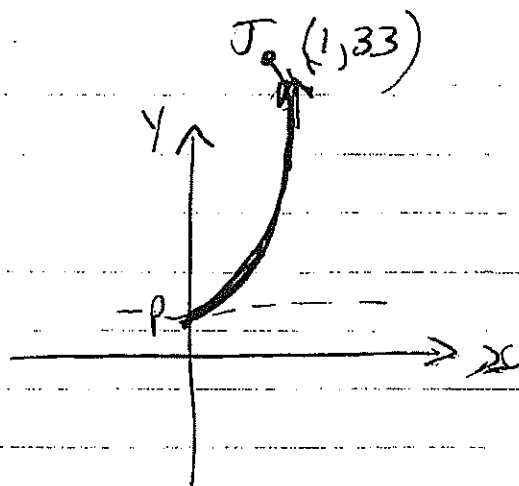
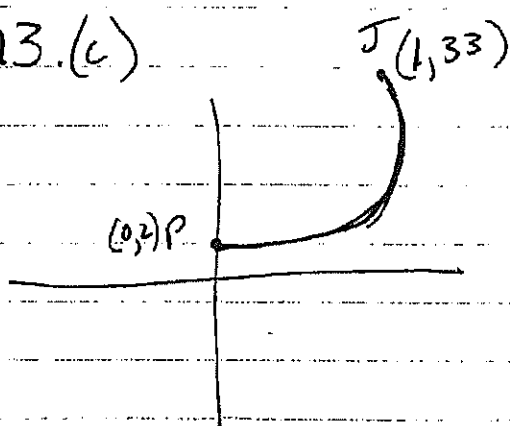
$$= 0 + \pi \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left(\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right)$$

$$= \frac{\pi^2}{2} \text{ cubic units.}$$

(3)

13. (c)



$$(i) \quad x = 2 \sin \frac{2t}{15} \quad y = 2 + \frac{5}{3}t - \frac{5}{3} \sin \left(\frac{t}{3} \right)$$

$$0 \leq t \leq \frac{15\pi}{2}$$

$$\text{At } t=0, \quad x=0, \quad y=2$$

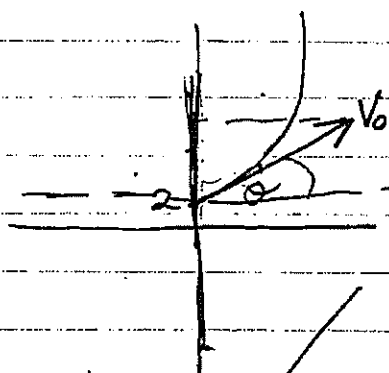
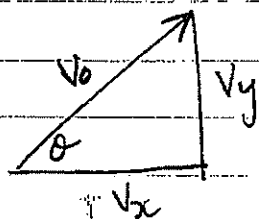
$$\Rightarrow \underline{P=(0, 2)} \quad \checkmark \quad (1)$$

$$(ii) \quad V_x = 2 \cos \frac{2t}{15} \times \frac{2}{15} \quad \text{and} \quad V_y = \frac{5}{3} - \frac{5}{9} \cos \frac{t}{3} \quad (2)$$

$$\underline{V_x = \frac{4}{15} \cos \frac{2t}{15}} \quad (1) \quad \checkmark$$

$$(2)$$

(iii)



Velocity components

$$V_x = V \cos \theta$$

$$V_y = V \sin \theta$$

$$\text{At } t=0, \quad V_x = \frac{4}{15}, \quad V_y = \frac{10}{9}$$

$$\text{Then } \tan \theta = \frac{V_y}{V_x}$$

$$= \frac{10}{9} \times \frac{15}{4}$$

$$\tan \theta = \frac{25}{6}$$

$$\theta = 1.33 \text{ radians}$$

$$\text{or } 76.5^\circ$$

Then angle made with forward direction

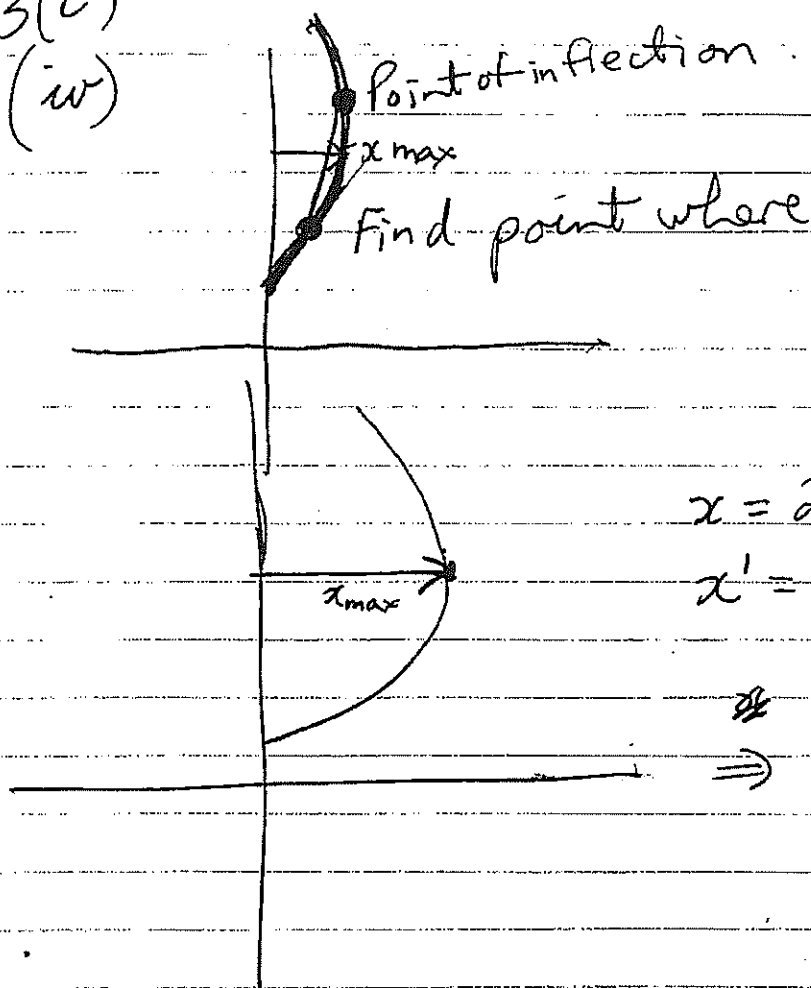
$$= 90 - 76.5$$

$$= 13.5^\circ \quad \checkmark$$

(2)

13(c)
(iv)

*



$$x = 2 \sin\left(\frac{2t}{15}\right)$$

$$x' = \frac{4}{15} \cos \frac{2t}{15} = 0 \text{ for stat pts}$$

$$\Rightarrow \cos \frac{2t}{15} = 0 \quad \sqrt{\frac{1}{2}}$$

$$\frac{2t}{15} = \frac{\pi}{2}, \frac{3\pi}{2} \quad \sqrt{\frac{1}{2}}$$

$$t = \frac{15\pi}{4}, \frac{45\pi}{4}$$

$t = 11.78 \text{ seconds}$ too big
 $\therefore$ at $t = 11.8 \text{ seconds}$ the ball starts to swing towards the jack.

2

13(c)

$$0 < t < \frac{15\pi}{2}$$

$$(v) \quad x = 2 \sin \frac{2t}{15}, \quad y = 2 + \frac{5}{3}t - \frac{5}{3} \sin \left(\frac{t}{3} \right)$$

Find t when $x=1$, $y=33$

$$\Rightarrow 1 = 2 \sin \frac{2t}{15} \quad (1) \quad \text{and} \quad 33 = 2 + \frac{5}{3}t - \frac{5}{3} \sin \left(\frac{t}{3} \right) \quad (2)$$

$$\sin \frac{2t}{15} = \frac{1}{2}$$

$$\frac{2t}{15} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$t = \frac{15\pi}{12}, \frac{75\pi}{12}, \frac{195\pi}{12} \text{ too big.}$$

$$t = \frac{5\pi}{4}, \frac{25\pi}{4}$$

$$31 = \frac{5}{3}t - \frac{5}{3} \sin \left(\frac{t}{3} \right)$$

Determine if values $t = \frac{15\pi}{12}$ or $\frac{75\pi}{12}$ satisfy (2).

$$t = \frac{15\pi}{12} \Rightarrow 31 = \frac{5}{3} \times \frac{15\pi}{12} - \frac{5}{3} \sin \left(\frac{15\pi}{36} \right) = 4.935 \times$$

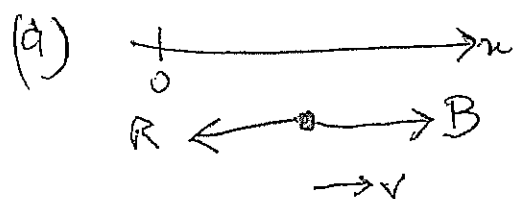
$$t = \frac{75\pi}{12} \Rightarrow 31 = \frac{5}{3} \times \frac{75\pi}{12} - \frac{5}{3} \sin \left(\frac{75\pi}{12} \right)$$

$$31 = 31.546 \times \text{ (close!)} \quad \checkmark$$

$\therefore$ Ball does not hit the Jack. $\checkmark$

(2)

Question 14



$$\begin{aligned} \text{Net force} &= B - R \\ &= 410 - 50 \\ &= 360 \text{ N} \end{aligned}$$

$$F = ma$$

$$\therefore 360 = 90a$$

$$a = 4 \text{ m/s}^2 \quad [2]$$

(b) Given $a = 4$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4$$

Integrate w.r.t. x :

$$\frac{1}{2} v^2 = 4x + C$$

$$v^2 = 8x + C'$$

Let $x=0$ when $t=0, v=0$

$$\therefore C' = 0$$

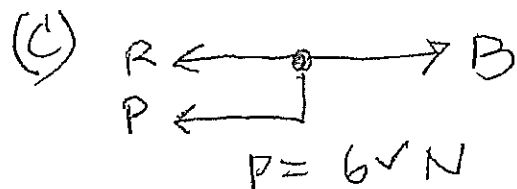
$$\therefore v^2 = 8x$$

When $x = 32$

$$\begin{aligned} v^2 &= 8 \times 32 \\ &= 256 \end{aligned}$$

$$\therefore v = 16 \text{ m/s } (v > 0)$$

[2]



$$P = 6v \text{ N}$$

$$F = ma$$

$$410 - 50 - 6v = 90a$$

$$a = \frac{360 - 6v}{90}$$

$$[1] \quad \therefore a = \frac{1}{15} (60 - v)$$

$$(d) \frac{dv}{dt} = \frac{1}{15} (60 - v)$$

$$\frac{dv}{60 - v} = \frac{1}{15} dt$$

Time from $v=16$ to $v=20$

$$\begin{aligned} t &= \int_{16}^{20} \frac{15}{60 - v} dv \\ &= 15 \left[-\ln(60 - v) \right]_{16}^{20} \\ &= -15 [\ln 40 - \ln 44] \\ &\approx 1.4 \text{ s} \end{aligned}$$

[3]

Q 14 (Cont'd)

(d) Horizontal $a = 0 \rightarrow \rightarrow$ Forces Boat: $500 \cos 30^\circ$ Resistance: -100 Parachute: $-T_2 \cos \theta$

$$\therefore 500 \cos 30^\circ - T_2 \cos \theta - 100 = 0 \quad \text{--- (1)}$$

Vertical $a = 0 \uparrow$ Forces Boat: $-500 \sin 30^\circ$ Weight: $-90g$ Parachute: $T_2 \sin \theta$

$$\therefore T_2 \sin \theta - 500 \sin 30^\circ - 90g = 0 \quad \text{--- (2)}$$

[3]

(e) From (1)

$$\begin{aligned} T_2 \cos \theta &= 500 \cos 30^\circ - 100 \\ &= 250\sqrt{3} - 100 \end{aligned}$$

From (2)

$$\begin{aligned} T_2 \sin \theta &= 500 \sin 30^\circ + 90g \\ &= 500 \times \frac{1}{2} + 900 \\ &= 1150 \end{aligned}$$

$$\begin{aligned} \therefore \tan \theta &= \frac{1150}{250\sqrt{3} - 100} \\ &= \frac{115}{25\sqrt{3} - 10} \end{aligned}$$

[2]

$$(g) T_1 = \frac{1150}{\sin \theta}$$

$$\text{from (f)} \quad \theta = 73.85^\circ$$

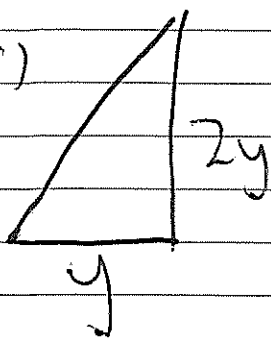
$$\therefore T_2 = 1197 \text{ N}$$

[2]

Q18

(a)

(i)



$$\text{Area} = \frac{1}{2} \times 2y \times y \\ = y^2.$$

$$\text{Since } x^2 + y^2 = r^2$$

$$\text{Area} = r^2 - x^2 \quad 2$$

(ii)

$$\delta V = (r^2 - x^2) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-r}^r (r^2 - x^2) \delta x.$$

$$= \int_{-r}^r r^2 - x^2 dx.$$

$$= 2 \int_0^r r^2 - x^2 dx$$

$$= 2 \left[xr^2 - \frac{x^3}{3} \right]_0^r$$

$$= 2 \left(r^3 - \frac{r^3}{3} \right)$$

$$= \frac{4r^3}{3}.$$

2

$$(b) (i) (a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0.$$

$$\frac{a^2 + b^2}{2} \geq ab.$$

$$\text{let } x = a^2 \Rightarrow a = \sqrt{x}$$

$$y = b^2 \Rightarrow b = \sqrt{y}.$$

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

2

$$(ii) \frac{\frac{1}{a} + \frac{1}{b}}{2} \geq \frac{1}{\sqrt{ab}}$$

$$\frac{a+b}{2ab} \geq \frac{1}{\sqrt{ab}}.$$

1

$$\frac{2ab}{a+b} \leq \sqrt{ab}.$$

$$(iii) \frac{x(x+1) + x^2 - 1 + x(x-1)}{x(x^2-1)} = \frac{3x^2 - 1}{x(x^2-1)}$$

$$= \frac{3x^2 - 3 + 2}{x(x^2-1)}$$

$$= \frac{3(x^2-1)}{x(x^2-1)} + \frac{2}{x(x^2-1)}$$

$$= \frac{3}{x} + \frac{2}{x(x^2-1)}$$

$$x^2 - 1 > 0 \text{ since } x > 1$$

$$\therefore x(x^2-1) > 0$$

$$\therefore \frac{2}{x(x^2-1)} > 0$$

$$\text{So } \frac{1}{x-1} + \frac{1}{x} + \frac{1}{x+1} > \frac{3}{x}$$

$$\begin{aligned} \text{(iv)} \quad H &= 1 + \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7}\right) + \left(\frac{1}{8} + \frac{1}{9} + \frac{1}{10}\right) + \dots + \left(\frac{1}{n-2} + \frac{1}{n-1} + \frac{1}{n}\right) \\ &= 1 + \frac{3}{3} + \frac{3}{6} + \frac{3}{9} + \dots + \frac{3}{n-1} \quad \text{let } n-1 = 3k. \\ &= 1 + 1 + \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) + \dots + \frac{1}{k}. \\ &= 1 + 1 + 1 + \dots + \frac{3}{k-1} \quad \text{let } k-1 = 3m. \\ &\geq 1 + 1 + 1 + \dots + \frac{1}{m}. \end{aligned}$$

As $n \rightarrow \infty$ this process can be continued

Thus $H = 1 + 1 + 1 + 1 + \dots + 1 + \dots$

$$\therefore \lim_{n \rightarrow \infty} H < \infty$$

(c) (i) Since w is a solution of

$$z^3 = 1$$

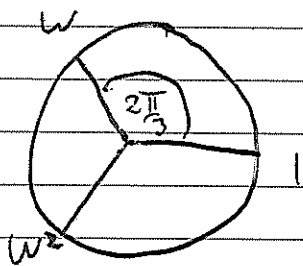
$$\therefore w^3 - 1 = 0.$$

$$(w-1)(w^2+w+1)=0$$

Since w is a non-real root
 $w \neq 1$.

$$\therefore w^2 + w + 1 = 0.$$

(ii)



$$\arg(w) = \frac{2\pi}{3}$$

$$\text{Given that } 1 + w + w^2 = 0$$

$$\text{SO } 1 + w = -w^2$$

$$1 + w^2 = -w$$

$$\text{LHS} = \left(\frac{w}{-w^2} \right)^k + \left(\frac{w^2}{-w} \right)^k$$

$$= (-1)^k (w^{-k}) + (-1)^k (w^k)$$

$$= (-1)^k (w^{-k} + w^k)$$

$$= (-1)^k 2 \cos\left(\frac{2\pi}{3}k\right)$$

$$\text{Given } z^n + z^{-n} = 2 \cos n\theta$$

Q16.

$$\begin{aligned}
 (a) \quad (i) \quad I_n &= \int_0^a (a-x)^n \cos x \, dx. \quad \dots \quad (A) \\
 &= \int_0^a (a-x)^n \frac{d}{dx} (\sin x) \, dx \\
 &= \left[(a-x)^n \sin x \right]_0^a - n \int_0^a (a-x)^{n-1} \sin x \, dx \\
 &= 0 - n \int_0^a (a-x)^{n-1} \sin x \, dx \\
 &= -n \int_0^a (a-x)^{n-1} \frac{d}{dx} (-\cos x) \, dx \\
 &= \left[-n (a-x)^{n-1} (-\cos x) \right]_0^a + n(n-1) \int_0^a (a-x)^{n-2} (-\cos x) \, dx \\
 &= (0 + n a^{n-1}) - n(n-1) \int_0^a (a-x)^{n-2} \cos x \, dx \\
 \therefore \quad \boxed{I_n = n a^{n-1} - n(n-1) I_{n-2}} \quad \text{NB } n \geq 2 \quad (B)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad I_3 &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x \right)^3 \cos x \, dx. \\
 &= 3 \left(\frac{\pi}{2} \right)^2 - 6 I_1 = \quad (C)
 \end{aligned}$$

$$\begin{aligned}
 \text{now } I_1 &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \cos x \, dx. \quad \text{from (A)} \\
 &= \int_0^{\frac{\pi}{2}} \frac{\pi}{2} \cos x \, dx - \int_0^{\frac{\pi}{2}} x \cos x \, dx \quad [\text{Don't use (B)}] \\
 &= \left[\frac{\pi}{2} \sin x \right]_0^{\frac{\pi}{2}} - \left[x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, dx \\
 &= \frac{\pi}{2} - \left[\frac{\pi}{2} - 0 - [\cos x]_0^{\frac{\pi}{2}} \right] \\
 &= \frac{\pi}{2} - \left(\frac{\pi}{2} - 1 \right) \\
 &= 1 \\
 \therefore \quad \boxed{I_3 = 3 \frac{\pi^2}{4} - 6} \quad \text{from (C)}
 \end{aligned}$$

Q16. (CONTD)

(b) (i) $BCQR$ is a cyclic quadrilateral

$$\angle RCQ = \angle RBQ \quad (90^\circ \text{ altitudes})$$

ie. angles are equal subtended by the chord RQ .

$$\therefore \angle BRQ = \angle BQC \quad (\text{angles in the same segment are equal})$$

$$\text{ie } \angle xRB = \theta.$$

$$(ii) \quad \angle xCQ = \angle xAQ = 90^\circ \quad (\text{data})$$

$\therefore xAQ \& C$ are concyclic (opposite angles supplementary)

$$(iii) \quad \angle xQC = \angle xAC = \theta \quad (\text{angles in the same segment standing on the same arc are equal})$$

(iv) $BxAR$ is a cyclic quadrilateral

$$\angle xBR = \angle xAR = 90^\circ \quad (\text{data})$$

(opposite angles supplementary)

$$\therefore \angle xRB = \angle xAB = \theta \quad (\text{angles in the same segment standing on same arc are equal})$$

$$\therefore \angle xAB = \angle xAC = \theta$$

$\therefore$ altitude PA bisects $\angle BAC$.

Similarly for the other angles.

Q16(CONTD)

(C) (i) Place the 41 0's

Choose the gaps and the end places

to place the 1's. (there are
42 spaces)

ie. $\binom{42}{9}$ ways to place the 1's.

(ii) Same logic as the above.

ie place the nine consecutive cards
into the 42 spaces such that no
two consecutive cards are adjacent

ie $\binom{42}{9}$.